2.58 Projectiles

Worked Solutions

1 a)

when $t = .5$

	x	y
s		
u	$25\cos 30°$	$25\sin 30°$
v	?	?
a	0	-9.8
t	.5	.5

$v = u + at$

$\mathbf{v} = (25\cos 30°)\mathbf{i} + (25\sin 30° - 9.8 \times .5)\mathbf{j}$

$= 21.7\mathbf{i} + 7.6\mathbf{j}$ ms^{-1} (3 SF)

$= 22\mathbf{i} + 7.6\mathbf{j}$ ms^{-1} (2 SF – AQA)

1 b)

Consider vertical components:

greatest height when $v = 0$

	x	y
s		?
u	$25\cos 30°$	$25\sin 30°$
v		0
a	0	-9.8
t		

$v^2 = u^2 + 2as$

$0^2 = (25\sin 30°)^2 + 2 \times -9.8 \times s$

$s = \frac{3125}{392} = 7.97$ m (3 SF)

8.0 m (2 SF – AQA)

1 c)

Consider vertical components:

time hits ground is when $s = 0$

	x	y
s		0
u	$25\cos 30°$	$25\sin 30°$
v		
a	0	-9.8
t		?

$s = ut + \frac{1}{2}at^2$

$0 = 25\sin 30° \times t + \frac{1}{2} \times -9.8 \times t^2$

$t = 0, \frac{125}{49}$

$t = 2.55$ seconds (3 SF)

$t = 2.6$ seconds (2 SF – AQA)

1 d)

Consider horizontal components:

when $t = \frac{125}{49}$

	x	y
s	?	
u	$25\cos 30°$	$25\sin 30°$
v		
a	0	-9.8
t	$\frac{125}{49}$	

$s = 25\cos 30° \times \frac{125}{49} + 0 = 55.2$ m (3 SF)

$s = 55$ m (2 SF – AQA)

2 a)

when $t = .4$

	x	y
s		
u	$10\cos 30°$	$10\sin 30°$
v	?	?
a	0	-9.8
t	.4	.4

$v = u + at$

$\mathbf{v} = (10\cos 30°)\mathbf{i} + (10\sin 30° - 9.8 \times .4)\mathbf{j}$

$= 8.66\mathbf{i} + 1.08\mathbf{j}$ ms^{-1} (3 SF)

$= 8.7\mathbf{i} + 1.1\mathbf{j}$ ms^{-1} (2 SF – AQA)

2 b)

Consider vertical components:

greatest height when $v = 0$

	x	y
s		?
u	$10\cos 30°$	$10\sin 30°$
v		0
a	0	-9.8
t		

$v^2 = u^2 + 2as$

$0^2 = (10\sin 30°)^2 + 2 \times -9.8 \times s$

$s = \frac{125}{98} = 1.28$ m (3 SF)

1.3 m (2 SF – AQA)

2 c)

Consider vertical components:

time hits ground is when $s = 0$

	x	y
s		0
u	$10\cos 30°$	$10\sin 30°$
v		
a	0	-9.8
t		?

$s = ut + \frac{1}{2}at^2$

$0 = 10\sin 30° \times t + \frac{1}{2} \times -9.8 \times t^2$

$t = 0, \frac{50}{49}$

$t = 1.02$ seconds (3 SF)

$t = 1.0$ seconds (2 SF – AQA)

2 d)

Consider horizontal components:

when $t = \frac{50}{49}$

	x	y
s	?	
u	$10\cos 30°$	$10\sin 30°$
v		
a	0	-9.8
t	$\frac{50}{49}$	

$s = 10\cos 30° \times \frac{50}{49} + 0 = 8.84$ m (3 SF)

$s = 8.9$ m (2 SF – AQA)

3 a)

when $t = 3$

	x	y
s		
u	$25\cos 45°$	$25\sin 45°$
v	?	?
a	0	-9.8
t	3	3

$v = u + at$

$\mathbf{v} = (25\cos 45°)\mathbf{i} + (25\sin 45° - 9.8 \times 3)\mathbf{j}$

$= 17.7\mathbf{i} - 11.7\mathbf{j}$ ms^{-1} (3 SF)

$= 18\mathbf{i} - 12\mathbf{j}$ ms^{-1} (2 SF – AQA)

3 b)

Consider vertical components:

greatest height when $v = 0$

	x	y
s		?
u	$25\cos 45°$	$25\sin 45°$
v		0
a	0	-9.8
t		

$v^2 = u^2 + 2as$

$0^2 = (25\sin 45°)^2 + 2 \times -9.8 \times s$

$s = \frac{3125}{196} = 15.9$ m (3 SF)

16 m (2 SF – AQA)

3 c)

Consider vertical components:

time hits ground is when $s = 0$

	x	y
s		0
u	$25\cos 45°$	$25\sin 45°$
v		
a	0	-9.8
t		?

$s = ut + \frac{1}{2}at^2$

$0 = 25\sin 45° \times t + \frac{1}{2} \times -9.8 \times t^2$

$t = 0, 3.607687659\ldots$

$t = 3.61$ seconds (3 SF)

$t = 3.7$ seconds (2 SF – AQA)

3 d)

Consider horizontal components:

when $t = 3.60768\ldots$

	x	y
s	?	
u	$25\cos 45°$	$25\sin 45°$
v		
a	0	-9.8
t	$3.60768\ldots$	

$s = 25\cos 45° \times 3.60768\ldots + 0 = 63.8$ m (3 SF)

$s = 64$ m (2 SF – AQA)

4 a)

	x	y
s		
u	$20\cos 20°$	$20\sin 20°$
v		
a	0	-9.8
t		

$\mathbf{u} = 20\cos 20°\mathbf{i} + 20\sin 20°\mathbf{j}$

$\mathbf{u} = 18.8\mathbf{i} + 6.84\mathbf{j}$ ms^{-1} (3 SF)

$\mathbf{u} = 19\mathbf{i} + 6.9\mathbf{j}$ ms^{-1} (2 SF – AQA)

4 b)

Time spent travelling is found when the ball has been displaced 10 metres vertically. Consider vertical components:

	x	y
s		-10
u	$20\cos 20°$	$20\sin 20°$
v		
a	0	-9.8
t		?

$s = ut + \frac{1}{2}at^2$

$-10 = 20\sin 20° \times t + \frac{1}{2} \times -9.8t^2$

$t = 2.287975366\ldots$ or $t = -0.8919747812$

$\therefore t = 2.28$ seconds (3 SF)

$t = 2.3$ seconds (2F – AQA)

4 c)

Since $t = 2.2879753666\ldots$

	x	y
s		-10
u	$20\cos 20°$	$20\sin 20°$
v	?	?
a	0	-9.8
t	$2.287975366\ldots$	$2.287975366\ldots$

$v^2 = u^2 + 2as$

$v^2 = (20\sin 20°)^2 + 2 \times -9.8 \times -10$

$v = \pm 15.5875\ldots$

$v = u + at = 20\cos 20° + 0 = 18.793852\ldots$

$\mathbf{v} = 18.8\mathbf{i} - 15.6\mathbf{j}$ ms^{-1} (3 SF)

$= 19\mathbf{i} - 16\mathbf{j}$ ms^{-1} (2 SF – AQA)

5 a)

$\tan\alpha = \frac{4}{3} \Rightarrow \sin\alpha = \frac{4}{5}, \cos\alpha = \frac{3}{5}$

Consider vertical components:

	x	y
s	75	-80
u	6	8
v		
a	0	-9.8
t		?

$-80 = 8t + \frac{1}{2} \times -9.8t^2$

$t = 4.938573444\ldots, -3.305920383\ldots$

$t = 4.93$ s (3 SF)

$t = 4.9$ s (2 SF – AQA)

5 b)

	x	y
s	x	$y-80$
u	6	8
v		
a	0	-9.8
t	t	t

$y - 80 = 8t - 4.9t^2$

$x = 6t \Rightarrow t = \frac{1}{6}x$

$y = 80 + 8(\frac{1}{6}x) - 4.9(\frac{1}{6}x)^2 = 80 + \frac{4}{3}x - \frac{49}{360}x^2$

5 c)

$y = 80 + \frac{4}{3}x - \frac{49}{360}x^2 = -\frac{49}{360}(x - \frac{240}{49})^2 + \frac{4080}{49}$

$\therefore$ max height of $\frac{4080}{49} = 83.3$ m (3SF)

83 m (2SF – AQA)

ALTERNATIVELY:

$\frac{dy}{dx} = \frac{4}{3} - \frac{49}{180}x$

Max when $\frac{dy}{dx} = 0$ so $x = \frac{240}{49}$

$\therefore$ max height of $\frac{4080}{49} = 83.3$ m (3SF)

83 m (2SF – AQA)

ALTERNATIVELY: Consider vertical components:

greatest height when $v = 0$

	x	y
s		?
u	6	8
v		0
a	0	-9.8
t		

$v^2 = u^2 + 2as$

$0^2 = 8^2 + 2 \times -9.8s \Rightarrow s = \frac{160}{49}$

Height above ground:

$\frac{160}{49} + 80 = 83.3$ m (3SF)

83 m (2SF – AQA)

6)

Consider the height of the ball when it has travelled a horizontal distance of 50 metres:

	x	y
s	50	
u	$22\cos 35°$	$22\sin 35°$
v		
a	0	-9.8
t	?	

$s = ut + \frac{1}{2}at^2$

$50 = 22\cos 35° t$

$\therefore t = \frac{50}{22\cos 35°}$

	x	y
s	50	
u	$22\cos 35°$	$22\sin 35°$
v		
a	0	-9.8
t	$\frac{50}{22\cos 35°}$	$\frac{50}{22\cos 35°}$

$s = ut + \frac{1}{2}at^2$

$s = 22\sin 35° \times \frac{50}{22\cos 35°} + \frac{1}{2} \times -9.8 \times (\frac{50}{22\cos 35°})^2$

$s = -2.70808754924\ldots$ metres

Add on the initial height:

$-2.708\ldots + 4.5 = 1.79$ m (3 SF)

Since the wall is 1.5 metres high, the ball passes over the wall.

7 a)

Consider the height of the ball when it has travelled a horizontal distance of 10 metres:

	x	y
s	10	
u	18	0
v		
a	0	-9.8
t	?	

$s = ut + \frac{1}{2}at^2$

$10 = 18t$

$\therefore t = \frac{5}{9}$

	x	y
s	10	
u	18	0
v		
a	0	-9.8
t	$\frac{5}{9}$	$\frac{5}{9}$

$s = ut + \frac{1}{2}at^2$

$s = 0 \times \frac{5}{9} + \frac{1}{2} \times -9.8 \times (\frac{5}{9})^2$

$s = -1.512345679\ldots$ metres

Add on the initial height:

$-1.512\ldots + 2.5 = 0.988$ m (3 SF)

Since the net is 1.2 metres high, the ball hits the net.

7 b)

Modelling the ball as a particle (dimensions of ball not allowed for, no spin on ball), no air resistance.

8)

$\tan\alpha = \frac{5}{12} \Rightarrow \sin\alpha = \frac{5}{13}, \cos\alpha = \frac{12}{13}$

27 m above ground is 2 m above start point

	x	y
s		2
u	$\frac{216}{13}$	$\frac{90}{13}$
v		
a	0	-9.8
t		?

$s = ut + \frac{1}{2}at^2$

$2 = \frac{90}{13}t + \frac{1}{2} \times -9.8t^2$

$t = 1.007914697\ldots, 0.4049581442\ldots$

$1.00791\ldots - 0.40495\ldots = 0.6029565528\ldots$

0.603 seconds (3 SF)

0.60 seconds (2 SF – AQA)

9 a)

$\tan\alpha = \frac{3}{4} \Rightarrow \sin\alpha = \frac{3}{5}, \cos\alpha = \frac{4}{5}$

	x	y
s	50	-40
u	$\frac{4}{5}u$	$\frac{3}{5}u$
v		
a	0	-9.8
t	t	t

$s = ut + \frac{1}{2}at^2$

Horizontal: $50 = \frac{4}{5}ut \Rightarrow ut = \frac{125}{2}$

Vertical: $-40 = \frac{3}{5}ut - 4.9t^2$

$-40 = \frac{3}{5} \times \frac{125}{2} - 4.9t^2 \Rightarrow t = \frac{5}{7}\sqrt{31}$

9 b)

$ut = \frac{125}{2} \Rightarrow u \times \frac{5}{7}\sqrt{31} = \frac{125}{2} \Rightarrow u = \frac{175}{2\sqrt{31}}$

9 c)

That g is exactly 9.8 ms^{-2}, there are no resistances to motion (like air resistance or wind), the dimension and shape of stone is ignored, any spin on the stone is ignored.

9 d)

It would not change the results as the modelling assumptions were that the original stone was modelled as a particle.

10)

$\tan\alpha = \frac{2}{5}$ gives $\sin\alpha = \frac{2}{\sqrt{29}}$ and $\cos\alpha = \frac{5}{\sqrt{29}}$

	x	y
s	x	$y-2$
u	$\frac{50}{\sqrt{29}}$	$\frac{-20}{\sqrt{29}}$
v		
a	0	-9.8
t	t	t

$x = \frac{50}{\sqrt{29}}t$ and $y - 2 = -\frac{20}{\sqrt{29}}t + \frac{1}{2} \times 9.8 \times t^2$

so $t = \frac{\sqrt{29}}{50}x$

$\Rightarrow y - 2 = -\frac{20}{\sqrt{29}}(\frac{\sqrt{29}}{50}x) + \frac{1}{2} \times -9.8 \times (\frac{\sqrt{29}}{50}x)^2$

$\Rightarrow y = 2 - \frac{2}{5}x - \frac{1}{2} \times 9.8 \times \frac{29}{2500}x^2$

$\Rightarrow y = 2 - \frac{2}{5}x - \frac{1421}{25000}x^2$ as req.

11 a)

$\tan\theta = \frac{3}{4} \Rightarrow \sin\theta = \frac{3}{5}, \cos\theta = \frac{4}{5}$

Need to know the vertical distance of the ball when the horizontal distance is 30 metres.

	x	y
s	35	
u	16	12
v		
a	0	-9.81
t	?	?

$s = ut + \frac{1}{2}at^2$

$36 = 16t + 0 \Rightarrow t = 2.25$

$s = 12(2.25) + \frac{1}{2} \times -9.81(2.25)^2 = 2.1684375$

$s = 2.17$ m (3 SF)

Yes the ball goes into the goal.

11 b)

No resistances to motion (air resistance / wind). Dimensions of ball and goal post not included so treating as a particle/rod. That g is 9.81 ms^{-2} exactly. No spin on the ball.

12 a)

$\tan\alpha = \frac{1}{10} \Rightarrow \sin\alpha = \frac{1}{\sqrt{101}}, \cos\alpha = \frac{10}{\sqrt{101}}$

	x	y
s		
u	$\frac{120}{\sqrt{101}}$	$\frac{-12}{\sqrt{101}}$
v		
a	0	-9.81
t	0.2	0.2

$v = u + at$

Horizontal:

$v = \frac{120}{\sqrt{101}} + 0 = 11.94044628\ldots$

Vertical:

$v = -\frac{12}{\sqrt{101}} + -9.81 \times 0.2 = -3.156044628\ldots$

$\tan^{-1}(\frac{3.15\ldots}{11.9\ldots}) = 14.8$

14.8° below the horizontal.

12 b)

	x	y
s		?
u	$\frac{120}{\sqrt{101}}$	$\frac{-12}{\sqrt{101}}$
v		
a	0	-9.81
t		0.4

$s = ut + \frac{1}{2}at^2$

$s = -\frac{12}{\sqrt{101}} \times 0.4 + \frac{1}{2} \times -9.81 \times (0.4)^2 = -1.262\ldots$

1.26 metres (3 SF)

12 c)

	x	y
s	?	
u	$\frac{120}{\sqrt{101}}$	$\frac{-12}{\sqrt{101}}$
v		
a	0	-9.81
t	0.4	

$s = ut + \frac{1}{2}at^2$

$s = \frac{120}{\sqrt{101}} \times 0.4 = 4.776178513\ldots$

4.77 metres (3 SF)

13 a)

The ball reduces in height by 1 metres.

	x	y
s	7	-1
u	$u\cos\theta$	$u\sin\theta$
v		
a	0	-10
t	2	2

$s = ut + \frac{1}{2}at^2$

$7 = 2u\cos\theta + 0 \Rightarrow u\cos\theta = \frac{7}{2}$

$-1 = 2u\sin\theta + \frac{1}{2} \times -10 \times 2^2 \Rightarrow u\sin\theta = \frac{19}{2}$

$u = \sqrt{(\frac{7}{2})^2 + (\frac{19}{2})^2} = \frac{1}{2}\sqrt{410} = 10.1\ \text{ms}^{-1}$ (3 SF)

$u = 10\ \text{ms}^{-1}$ (1 SF – AQA)

13 b)

$\frac{u\sin\theta}{u\cos\theta} = \tan\theta = \frac{\frac{19}{2}}{\frac{7}{2}} = \frac{19}{7} \Rightarrow \theta = 69.8°$ (3 SF)

$\theta = 70°$ (1 SF – AQA)

14 a)

	x	y
s		$+1$
u	$18\cos 30°$	$18\sin 30°$
v		
a	0	-10
t	t	t

$s = ut + \frac{1}{2}at^2$

Vertical:

$1 = 18\sin 30°t + \frac{1}{2} \times -10 \times t^2$

$t = \frac{9 \pm \sqrt{61}}{10}$ so $t = 1.68124968\ldots$

Horizontal:

$s = 18\cos 30° \times 0.7116805376 = 26.2$ m (3 SF)

$s = 30$ m (1 SF – AQA)

14 b)

Horizontal: $v = 18\cos 30°$

Vertical: $v^2 = (18\sin 30°)^2 + 2 \times -10 \times -2$

$\therefore v = 11$

$v = 9\sqrt{3}\mathbf{i} - 11\mathbf{j}$

$\tan^{-1}(\frac{11}{9\sqrt{3}}) = 35.2\ldots$

$35.2°$ below the horizontal

15)

Ball A

	x	y
s	d	h
u	$5\sqrt{3}$	5
v		
a	0	-10
t	t	t

Ball B

	x	y
s	d	h
u	20	$20\sqrt{3}$
v		
a	0	-10
t	$t-1$	$t-1$

$s = ut + \frac{1}{2}at^2$

Ball A:

$d = 5\sqrt{3}t$

$h = 5t - 5t^2$

Ball B:

$d = 20(t-1)$

$h = 20\sqrt{3}(t-1) - 5(t-1)^2$

Equate horizontal distances:

$5\sqrt{3}t = 20(t-1) \Rightarrow t = 1.76307\ldots$ seconds

When $t = 1.76307\ldots$

Height of ball A: $5(1.76307) - 5(1.76307)^2 = -6.734788798\ldots$

Height of ball B: $20\sqrt{3}(3 + 2\sqrt{3}) - 5(3 + 2\sqrt{3})^2 = 23.5393\ldots$

Difference in height of 30.3 m (3 SF)

30 m (1 SF – AQA)

16 a)

	x	y
s		$y - a$
u	$10\cos\theta$	$10\sin\theta$
v		
a	0	-10
t	t	t

$s = ut + \frac{1}{2}at^2$

Horizontal: $x = 10\cos\theta \times t$

Vertical: $y = 10\sin\theta \times t - 5t^2$

$t = \frac{x}{10\cos\theta}$

$y - a = 10\sin\theta(\frac{x}{10\cos\theta}) - 5(\frac{x}{10\cos\theta})^2$

$y = a + x\tan\theta - \frac{x^2}{20\cos^2\theta}$

16 b)

$0 = 0.7 + 9.2\tan\theta - \frac{9.2^2}{20\cos^2\theta}$

$0 = 0.7 + 9.2\tan\theta - 4.232\sec^2\theta$

$4.232\sec^2\theta - 9.2\tan\theta - 0.7 = 0$

$4.232(\tan^2\theta + 1) - 9.2\tan\theta - 0.7 = 0$

$4.232\tan^2\theta + 4.232 - 9.2\tan\theta - 0.7 = 0$

$4.232\tan^2\theta - 9.2\tan\theta + 3.532 = 0$

$\tan\theta = 1.675922486\ldots$ or

$\tan\theta = 0.4979905572\ldots$

$\theta = 59.2°$ or $\theta = 26.5°$ (3 SF)

$\theta = 60°$ or $\theta = 30°$ (1 SF)

So it was fired at $\theta = 59.2°$ or $\theta = 26.5°$ (3 SF) above the horizontal

17 a)

Ball A

	x	y
s	x	y
u	$10\cos 30°$	$10\sin 30°$
v		
a	0	$-g$
t	t	t

Ball B

	x	y
s	$5-x$	y
u	$u\cos 45°$	$u\sin 45°$
v		
a	0	$-g$
t	t	t

$s = ut + \frac{1}{2}at^2$

Same heights:

Ball A: $y = 10\sin 30° \times t + \frac{1}{2} \times -g \times t^2$

$y = 5t - \frac{1}{2}gt^2$

Ball B: $y = u\sin 45° \times t + \frac{1}{2} \times -g \times t^2$

$y = \frac{\sqrt{2}}{2}ut - \frac{1}{2}gt^2$

$5t - \frac{1}{2}gt^2 = \frac{\sqrt{2}}{2}ut - \frac{1}{2}gt^2 \Rightarrow 5 = \frac{\sqrt{2}}{2}u \Rightarrow u = 5\sqrt{2}$

17 b)

Equate distances:

$5\sqrt{3}t = -5\sqrt{2}\cos 45°t + 5$

$5\sqrt{3}t = -5t + 5$

$\sqrt{3}t + t = 1$

$t = \frac{1}{1+\sqrt{3}} = \frac{-1+\sqrt{3}}{2}$ seconds

17 c)

$x = 10\cos 30° \times (\frac{-1+\sqrt{3}}{2}) = \frac{15-5\sqrt{3}}{2}$

$\therefore 3.17$ m

18)

	x	y
s	x	y
u	$u\cos\theta$	$u\sin\theta$
v		
a	0	$-g$
t	t	t

$s = ut + \frac{1}{2}at^2$

Horizontal: $x = u\cos\theta\, t \Rightarrow t = \frac{x}{u\cos\theta}$

Vertical: $y = u\sin\theta\, t - \frac{1}{2}gt^2$

$y = u\sin\theta(\frac{x}{u\cos\theta}) - \frac{1}{2}g(\frac{x}{u\cos\theta})^2$

$y = (\frac{\sin\theta}{\cos\theta})x - \frac{1}{2}g(\frac{x^2}{u^2\cos^2\theta})$

$y = \tan\theta\, x - \frac{g}{2u^2\cos^2\theta}x^2 = \tan\theta\, x - \frac{g}{2u^2}\sec^2\theta\, x^2$ as req.

19 a)

maximum height occurs when $v = 0$ vertically:

	x	y
s		h
u	$u\cos\theta$	$u\sin\theta$
v		0
a	0	$-g$
t		

$v^2 = u^2 + 2as$

$0 = u^2\sin^2\theta - 2gh \Rightarrow 2gh = u^2\sin^2\theta$

$\Rightarrow h = \frac{u^2}{2g}\sin^2\theta$ as req.

19 b)

$h = \frac{10^2}{2g}\sin^2\theta = \frac{50}{g}\sin^2\theta$

But $0 \leq \sin^2\theta \leq 1$ so $0 \leq \frac{50}{g}\sin^2\theta \leq \frac{50}{g}$

$\therefore 0 \leq h \leq \frac{50}{g}$

19 c)

$5 = \frac{12^2}{2\times 10}\sin^2\theta \Rightarrow \sin^2\theta = \frac{25}{36} \Rightarrow \sin\theta = \pm\frac{5}{6}$

$\theta = 56.4°$

20 a)

Maximum distance occurs when $s = 0$ vertically:

	x	y
s	x	0
u	$u\cos\theta$	$u\sin\theta$
v		
a	0	$-g$
t	t	t

$s = ut + \frac{1}{2}at^2$

Vertical:

$0 = u\sin\theta\, t - \frac{1}{2}gt^2 = t(u\sin\theta - \frac{1}{2}gt)$

$\Rightarrow u\sin\theta - \frac{1}{2}gt = 0 \Rightarrow u\sin\theta = \frac{1}{2}gt \Rightarrow t = \frac{2u}{g}\sin\theta$

20 b)

Horizontal:

$x = u\cos\theta\, t$ so $x = u\cos\theta \times \frac{2u}{g}\sin\theta$

$= \frac{u^2}{g} \times 2\cos\theta\sin\theta = \frac{u^2}{g}\sin 2\theta$ as req.

20 c)

$20 = \frac{15^2}{10} \sin 2\theta \Rightarrow \sin 2\theta = \frac{8}{9}$

$\theta = 31.4^\circ$ or $\theta = 58.6^\circ$

$\therefore 31.4^\circ < \theta < 58.6^\circ$